

Technical Report

Capstone Project – Real World Business Analysis

1. Introduction

The increasing availability of data in modern organizations has created opportunities for businesses to make more informed, data-driven decisions. However, deriving meaningful insights from raw data requires structured analytical workflows that combine technical skills with business understanding. This project presents a complete end-to-end business analysis designed to demonstrate professional analytics practices using real-world datasets.

The aim of this capstone project is to perform a comprehensive analysis starting from raw data preparation and ending with actionable business recommendations. The project incorporates multiple datasets representing different business domains, including sales performance, customer churn behavior, and house price information. By integrating these datasets into a unified analytical workflow, the project simulates a realistic industry scenario where analysts must translate data into insights that influence business strategy.

The analysis process was intentionally structured into modular stages using separate notebooks to ensure clarity, reproducibility, and professional project organization. Each phase builds on the previous stage, demonstrating a systematic approach to data analysis.

2. Project Objectives

The primary objectives of this capstone project include:

- Demonstrate a complete end-to-end data analytics workflow.
 - Apply data cleaning and preprocessing techniques to raw datasets.
 - Perform exploratory data analysis (EDA) to understand patterns and relationships.
 - Use statistical and analytical methods to extract meaningful insights.
 - Create professional visualizations for interpretation and communication.
 - Translate technical analysis into actionable business recommendations.
 - Produce portfolio-ready documentation and reproducible analysis.
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3. Project Workflow Overview

The project follows a structured multi-phase workflow:

Phase 1 – Data Preparation

- Loading raw datasets
- Data inspection and validation
- Missing value handling
- Duplicate removal
- Feature engineering
- Exporting cleaned data

Phase 2 – Exploratory Data Analysis (EDA)

- Statistical summaries
- Distribution analysis
- Outlier detection
- Trend analysis
- Correlation exploration

Phase 3 – Advanced Analysis

- Business KPI computation
- Statistical comparison and validation
- Segment-level performance analysis
- Relationship analysis

Phase 4 – Final Business Analysis

- Insight consolidation
- Business recommendation development
- Implementation planning

This structure ensures that insights are derived through a repeatable and transparent process.

4. Dataset Description

The project utilizes three independent datasets, each representing a different analytical domain.

4.1 Sales Data

The sales dataset contains transactional-level information including pricing, quantity, and category-related variables. This dataset is used to evaluate business performance, analyze sales trends, and identify factors affecting revenue generation.

4.2 Customer Churn Data

The churn dataset includes customer attributes along with churn indicators representing whether customers continued or discontinued service. This dataset supports customer behavior analysis and helps identify patterns associated with retention risks.

4.3 House Prices Data

The house price dataset contains numerical and categorical features related to housing properties. The dataset provides an opportunity to explore feature relationships and distribution characteristics useful for comparative analysis.

All datasets contain sufficient records (500+ rows) to support meaningful statistical analysis and visualization.

5. Data Cleaning & Preparation

Data cleaning was performed using:

notebooks/1_data_cleaning.ipynb

5.1 Initial Inspection

Initial exploration included:

- Checking data dimensions
- Reviewing column data types
- Inspecting sample records
- Identifying missing values

5.2 Missing Value Treatment

Missing values were handled using a systematic strategy:

- Numerical columns → replaced with median values
- Categorical columns → replaced with mode values

This approach ensures robustness against outliers while preserving dataset distribution.

5.3 Duplicate Handling

Duplicate rows were detected and removed to avoid biased analysis and incorrect aggregation results.

5.4 Feature Engineering

Additional features were created where applicable to enhance analytical value, such as:

- Calculation of derived metrics (e.g., Total Sales = Quantity × Price)

5.5 Validation & Export

After cleaning, datasets were validated for consistency and saved as cleaned versions for downstream analysis.

6. Exploratory Data Analysis (EDA)

EDA was conducted in:
notebooks/2_eda.ipynb

The purpose of EDA was to understand data behavior before applying advanced analysis.

6.1 Distribution Analysis

Histograms and density plots were used to understand how variables are distributed across each dataset. This helped identify skewed distributions and potential anomalies.

6.2 Correlation Analysis

Correlation heatmaps were generated to evaluate relationships between numerical variables. Strong positive or negative relationships were identified, providing early indications of impactful features.

6.3 Outlier Detection

Boxplots were used to detect potential outliers that could influence statistical analysis or visualization interpretation.

6.4 Categorical Analysis

Count plots and categorical comparisons provided insights into customer distribution and churn patterns.

6.5 Relationship Analysis

Scatter plots helped visualize relationships between variables such as pricing and sales performance.

The EDA phase revealed patterns later confirmed in advanced analysis.

7. Advanced Analysis

Advanced analytics was implemented in:
notebooks/3_advanced_analysis.ipynb

7.1 Business KPI Analysis

Key performance indicators were calculated, including:

- Total revenue
- Average sales value
- Maximum transaction values
- Customer churn rate
- Average housing price

These KPIs provide measurable indicators of business performance.

7.2 Statistical Analysis

Statistical methods were applied to validate observations from EDA. Hypothesis testing was used to determine whether differences between groups were statistically significant.

7.3 Segment Analysis

Segment-level performance analysis allowed comparison across categories, highlighting areas with strong or weak performance.

7.4 Feature Relationship Analysis

Correlation-based analysis identified variables most strongly associated with business outcomes.

8. Visualizations

A variety of professional visualizations were developed throughout the analysis:

- Distribution Histograms
- Boxplots for outlier analysis
- Correlation Heatmaps
- Line charts for trend analysis
- Scatter plots for relationship analysis
- Bar charts for categorical comparison

All visualizations were designed to prioritize readability and business interpretation.

9. Key Technical Findings

The analysis produced several notable technical insights:

- Pricing-related variables exhibit a strong correlation with overall sales performance.
- Customer churn displays identifiable behavioral patterns suggesting segmentation opportunities.

- Statistical tests confirmed meaningful differences between analyzed groups.
- Certain variables consistently influenced performance metrics across datasets.

These findings demonstrate how structured analysis can uncover actionable relationships.

10. Business Recommendations

Based on technical findings, the following recommendations were proposed:

1. Adjust pricing strategies based on correlation-driven insights.
 2. Develop targeted retention programs focused on high-risk customers.
 3. Prioritize investment in high-performing business categories.
 4. Continuously monitor KPIs for proactive decision-making.
 5. Use segmentation-based strategies for personalized interventions.
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11. Implementation Roadmap

Month 1

- Deploy pricing optimization strategy
- Establish KPI monitoring dashboard

Month 2

- Launch customer retention initiatives
- Track early results

Month 3

- Evaluate impact and refine strategies
 - Continue data-driven monitoring
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12. Reproducibility & Code Organization

The project follows a modular structure:

```
notebooks/  
├── 1_data_cleaning.ipynb  
├── 2_eda.ipynb  
├── 3_advanced_analysis.ipynb  
└── capstone_analysis.ipynb
```

This separation improves readability, maintainability, and reproducibility.

13. Limitations

- Analysis limited by available features.
 - Additional external data could improve accuracy.
 - Predictive modeling was outside the scope of this project.
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14. Conclusion

This capstone project successfully demonstrates a complete analytical workflow from raw data preprocessing to advanced analysis and business recommendation development. The structured approach highlights both technical proficiency and business-oriented thinking, showcasing the ability to transform data into meaningful insights that support decision-making.

The final deliverables reflect industry-standard analytics practices and provide a portfolio-ready example of applied data analysis.